

ASSIGN Parent Model: Constrained Homophilic Bipartite Formation

Two established ideas supply the ingredients; the synthesis is our assignment mechanism.

Configuration-model logic

Give each team a fixed number of stubs. R_j is our notation for the number of unmatched team stubs remaining.

Quayle-style homophily

Similarity is a generative preference. Closer individual–team states receive greater attachment weight.

$R_{j,t}^{(s)} = C_j - n_{j,t}^{(s)}$
structural availability /
remaining capacity

$$P = (i \rightarrow j, t, s) = \frac{R_{j,t}^{(s)} e^{-\rho |A_i^* - \mu_{j,t-1}|}}{\sum_l R_{l,t}^{(s)} e^{-\rho |A_i^* - \mu_{l,t-1}|}}$$

our synthesis: capacity-preserving matching biased by homophily

$e^{-\rho |A_i^* - \mu_{j,t-1}|}$
non-structural
similarity attraction

After each assignment

If i joins team j :
 $R_{j,t}^{(s+1)} = R_{j,t}^{(s)} - 1$
Full team therefore has 0 remaining stubs.

After each period

Recalculate the team centroid $\mu_{j,t}$
from the new roster. That state can
guide the next formation period.

Dynamic extension

$\mu_{j,t-1} \rightarrow \text{ASSIGN at } t \rightarrow \text{roster at } t \rightarrow \mu_{j,t} \rightarrow \text{ASSIGN at } t+1$

Current Use: One-Shot Homophilic Initial-League Generator

We derive the smallest model needed for the paper—no longitudinal updating.

1 Fixed empirical team states

Initialize each team with an empirical centroid μ_j . These centroids remain fixed for the entire formation round.

2 Individual states

Use the observed / empirically informed A_i values for the complete player population.

3 All slots initially open

$R_j^{(0)} = C$ for every team j .
For NCAA MBB: $C = 15$.

4 Random player queue

Randomly permute players before sequential assignment; order is not based on A_i .

$$P = (i \rightarrow j | s) = \frac{R_j^{(s)} e^{-\rho |A_i - \mu_j|}}{\sum_l R_l^{(s)} e^{-\rho |A_i - \mu_l|}}$$

$R_j^{(0)} = C$ for all teams; after each assignment, the selected team's R_j decreases by one

i = the individual/player currently being assigned

j = a candidate team

A_i = individual i 's ability/performance value

μ_j = fixed empirical centroid for team j

ρ = strength of homophilic attraction

$R_j^{(s)}$ = remaining open roster positions (unmatched team stubs)
for team j at assignment step s

l = index over all teams with remaining capacity

s = the sequential assignment step within this one formation run

$\rho = 0 \rightarrow$ random baseline

$$P(i \rightarrow j | s) = R_j^{(s)} / \sum_l R_l^{(s)}$$

Uniform matching over all remaining team stubs.

$\rho > 0 \rightarrow$ homophilic bias

The same capacity-preserving process increasingly favors teams with μ_j close to A_i .

One run = one realization

Repeat from the same initial conditions to obtain an ensemble of possible leagues—not longitudinal iterations.

The combined probability rule is our synthesis; Fosdick supports the stub/configuration logic and Quayle supports the similarity-weighted homophily logic.